Tennis (Group 3)

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Table of contents

Introduction

Tennis is a globally popular sport renowned for its mix of physical agility, mental strategy, and endurance. Matches are structured into points, which accumulate to form games; games form sets, and winning the required number of sets secures the match. Each point carries significant weight, with momentum often shifting rapidly on the court. Analyzing match data such as serve percentages and return success offers valuable insights into player performance. These statistics are indispensable tools for understanding the strengths, weaknesses, and unique playing styles, particularly when examining the competitive landscape of 2017 men's singles tennis.

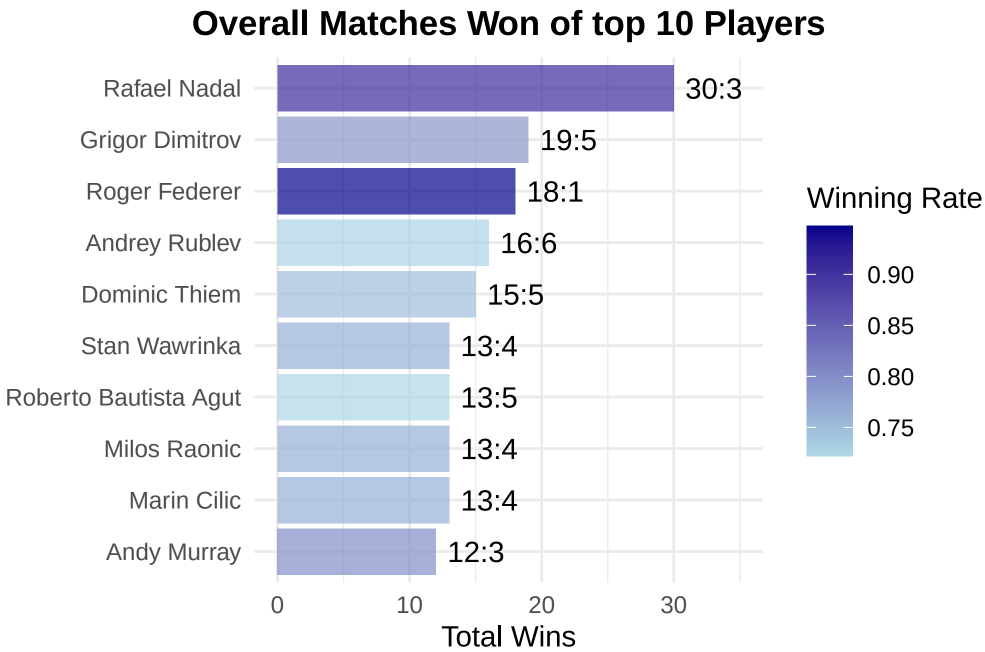
Describing the Data

Structure of the Data Set

The Tennis dataset contains the results of 925 men's singles tennis matches in the year 2017 for 4 Grand Slam Tournaments (Australian Open, Roland Garros, US Open, Wimbledon) and 2 ATP Tours (Beijing, Brisbane). Each match includes the tournament and match information, winning and losing players, sets and games won by the players, and serving/returning statistics of both winning and losing players.

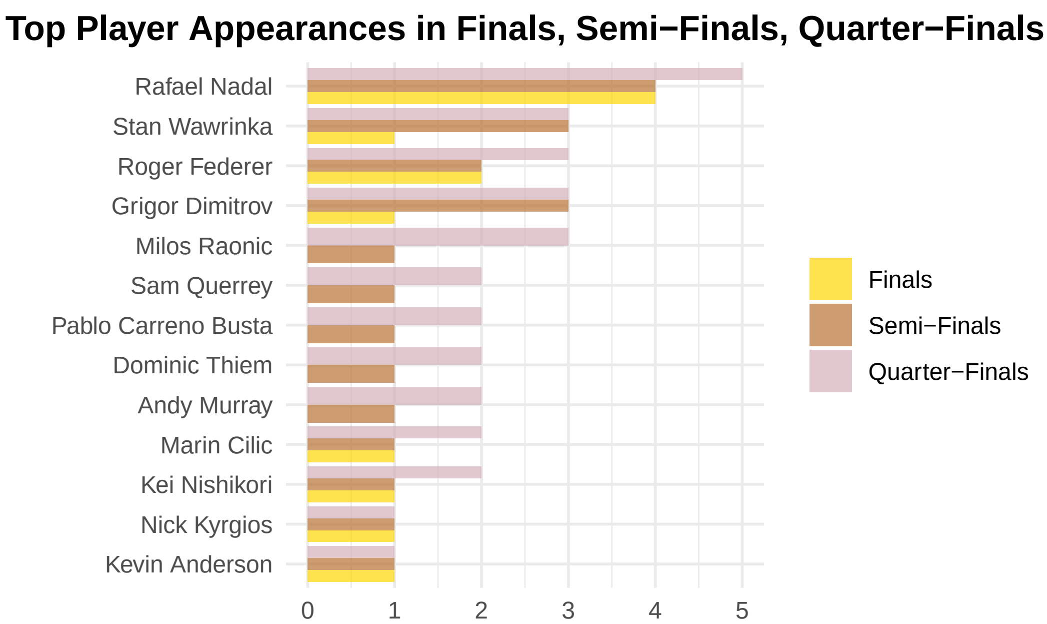
Key findings

One of the primary objectives in analyzing this dataset was to explore the question: *What makes a great tennis player?* To address this, we adopted two distinct perspectives. The first focused on match outcomes, aiming to identify which players had the highest number of match victories.



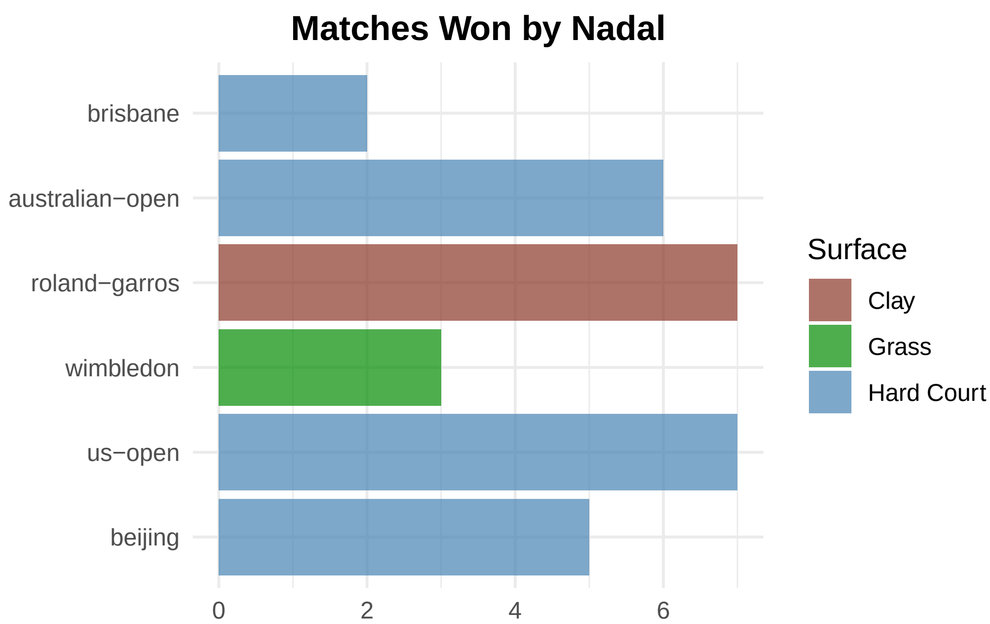
This plot displays the top 10 tennis players with the most match wins. Each bar represents a player's total number of wins, with the bars ordered from most to least wins. The fill color of each bar reflects the player's win-loss ratio—darker blue indicates a higher winning rate. A label on each bar shows the win-loss record (e.g., 30:5). This visualization helps highlight not only who won the most matches but also how dominant they were in terms of win percentage. It is immediately apparent that players such as Nadal and Federer demonstrated a high level of consistency across all recorded matches in the dataset.

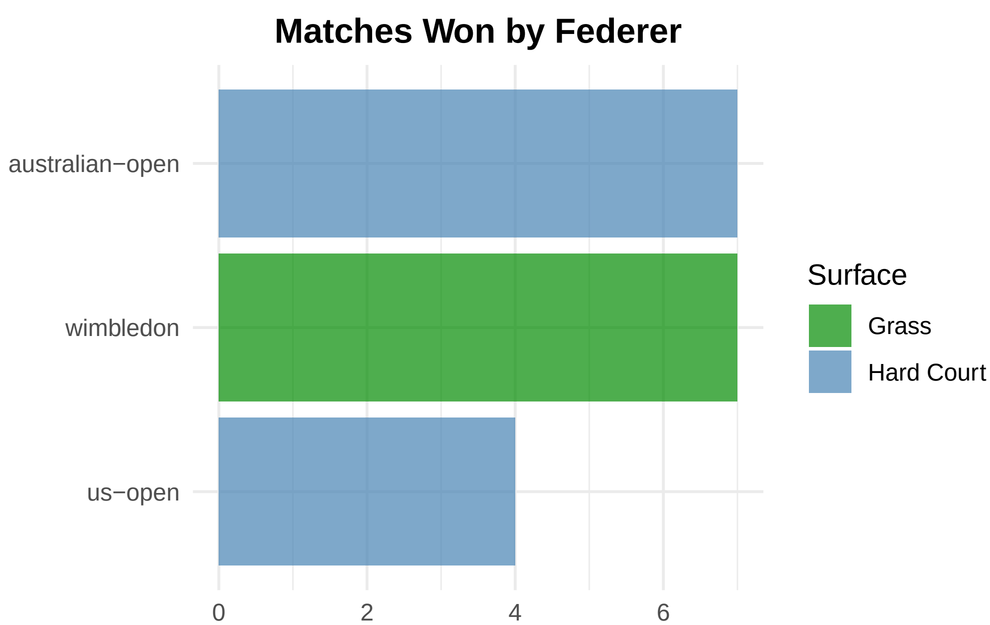
Another way to identify the top-performing players is by examining their appearances in the final rounds of the six tournaments under analysis. The following plot demonstrates that, by this measure as well, the same names consistently emerge as the leading players.



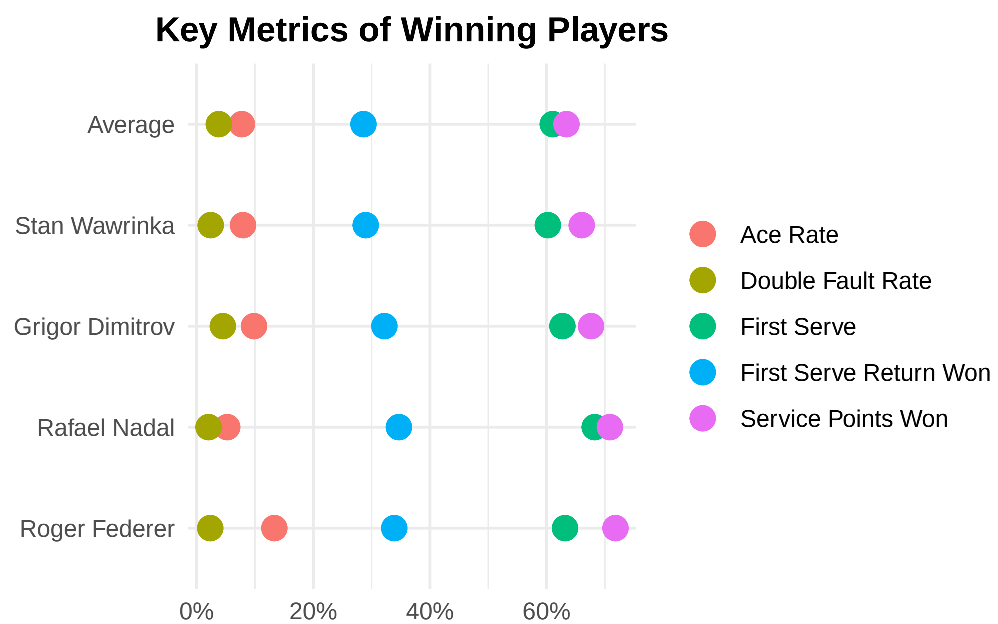
Again, Nadal and Federer are among the best performing players, having reached the most finals.

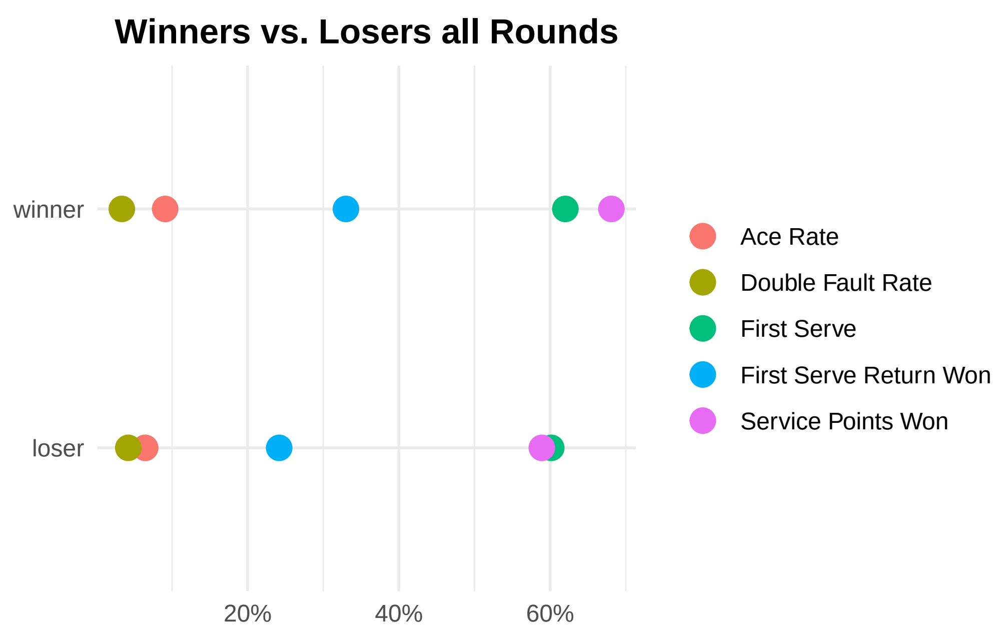
We also explored whether different court surfaces may have influenced match outcomes, considering that some players perform better on clay, while others excel on grass or hard courts. To illustrate this, we focused again on Nadal and Federer, two of the top-performing players in our dataset. The analysis shows that Nadal dominated on clay courts, whereas Federer secured many of his wins on grass. Both players performed strongly on hard courts. However, while these patterns highlight their individual strengths, they do not offer a clear explanation for *why* these players were so consistently successful overall.





The second perspective was more analytical, involving a comparison of performance metrics between players. The goal was to determine which factors distinguish successful players from those who are less successful. To achieve this, we calculated and compared several key performance indicators, including ace rate, double fault rate, first serve percentage, first serve return points won, and total service points won. Each of these metrics was expressed as a percentage, allowing for a clear and effective visual comparison across the two groups.





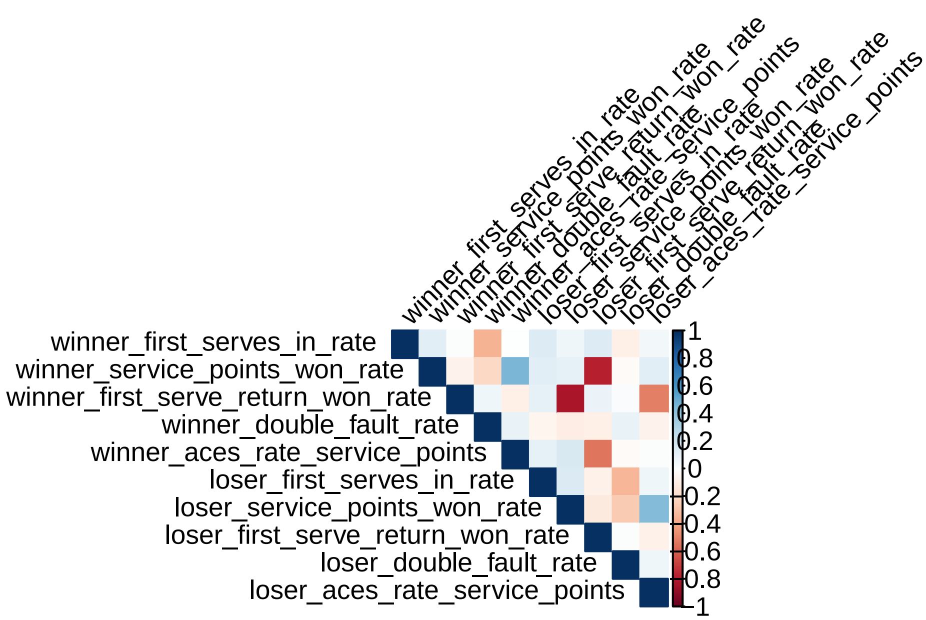
From this analysis, we conclude that metrics such as first serve percentage, service points won, and first serve return success are more indicative of a winning player than ace rate or double fault rate. This suggests that strong serving ability is a fundamental factor in overall player success. This outcome aligns with general tennis knowledge - serving is the only aspect of the game entirely within a player's control, unaffected by the opponent's actions.

It is worth acknowledging that these findings may appear somewhat self-evident. Our analysis ultimately confirms that players with effective serves and solid return games are more likely to winan - insight already familiar to most tennis players. However, the scope of our conclusions was limited by the dataset, which lacked additional informative variables such as player age or break point statistics. As a result, we were constrained to focus on the available performance metrics.

A detailied summary of all attempted plots can be found here in this shinyapp.

Statistical Analysis

Correlation



The corrplot serves as a graphical representation of a correlation matrix, offering valuable insights into the relationships among multiple performance metrics in tennis matches. In this case, the upper triangle of the plot visualizes correlations among winner-specific variables, while the lower triangle reflects relationships among corresponding loser-specific variables. The color scheme conveys both the direction and strength of each pairwise Pearson correlation coefficient: dark blue indicates strong positive correlations (close to +1), dark red represents strong negative correlations (close to –1), and white or pale shades signify weak or negligible correlations (values near 0).

Among match winners, several serve-related variables show strong positive connections. For example, the first serve in rate is closely linked to both the percentage of service points won and the number of aces per service point. This suggests that players who consistently land their first serves tend to perform better in service situations and hit more aces. These relationships reflect a cohesive serve performance, where both precision and power contribute jointly to match success.

Among match losers, the same serve-related correlations exist but are notably weaker. This indicates that losing players often show less consistent or effective serve strategies. Their performance lacks the strong connections seen among the winners, hinting at serve quality as a potential factor in match outcomes.

The double fault rate stands out as a negatively correlated variable across both groups. For winners and losers alike, it shows inverse relationships with important performance indicators such as service points won. This implies that committing more double faults is statistically harmful to service effectiveness. The negative correlation confirms that reducing unforced service errors plays a key role in staying competitive, especially on serve.

Return performance presents a different pattern. The first serve return won rate shows only moderate correlations with other match statistics, particularly among winners. This means that a player's ability to return serves effectively tends to operate independently from other areas of performance, like serving strength or ace production. This independence reflects how successful players may use different winning strategies some excel through powerful serves, while others rely on strong returns. In short, return skills contribute uniquely and do not necessarily align with patterns seen in serve metrics.

Hypothesis testing

The difference in the performance of winners and losers were compared using tests of hypothesis. This is done under the assumption that the matches in the dataset are considered "representative". Both mean and median were used to see if the results change based on the distribution of the dataset.

The table below shows that winning players have a significantly better performance than losing players across all player statistics, which is expected.

Player Statistics	**Mean Difference (Winner – Loser)**	**Median Difference (Winner – Loser)**
Service Ace (%)	0.03*[1]	0.02*

[1] An * means significant at 5% alpha.

Double Faults (%)	-0.01*	-0.01*
First Serve In (%)	0.02*	0.02*
Service Points Won (%)	0.11*	0.10*
First Serve Return Won (%)	0.11*	0.10*
Tiebreaks Won (%)	0.57*	1*

Difference in Performance Between Winning and Losing Players

Regression Model

For modeling we first considered the following model: win ~ aces_rate + double_faults_rate + first_serve_pct + service_win_pct + return_win_pct. The results were as follows:

```
Initial model summary



Call:
glm(formula = win ~ aces_rate + double_faults_rate + first_serve_pct +
    service_win_pct + return_win_pct, family = "binomial", data = train)

Coefficients:
                   Estimate Std. Error z value Pr(>|z|)
(Intercept)        -30.4042     1.9562 -15.543   <2e-16 ***
aces_rate           -3.7386     1.9089  -1.959   0.0502 .
double_faults_rate  -0.9907     4.3683  -0.227   0.8206
first_serve_pct     -0.6706     1.3697  -0.490   0.6244
service_win_pct     38.3055     2.4297  15.766   <2e-16 ***
return_win_pct      23.9702     1.6248  14.753   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 2051.71  on 1479  degrees of freedom
Residual deviance:  779.04  on 1474  degrees of freedom
AIC: 791.04

Number of Fisher Scoring iterations: 7

VIF values for the initial model:
```

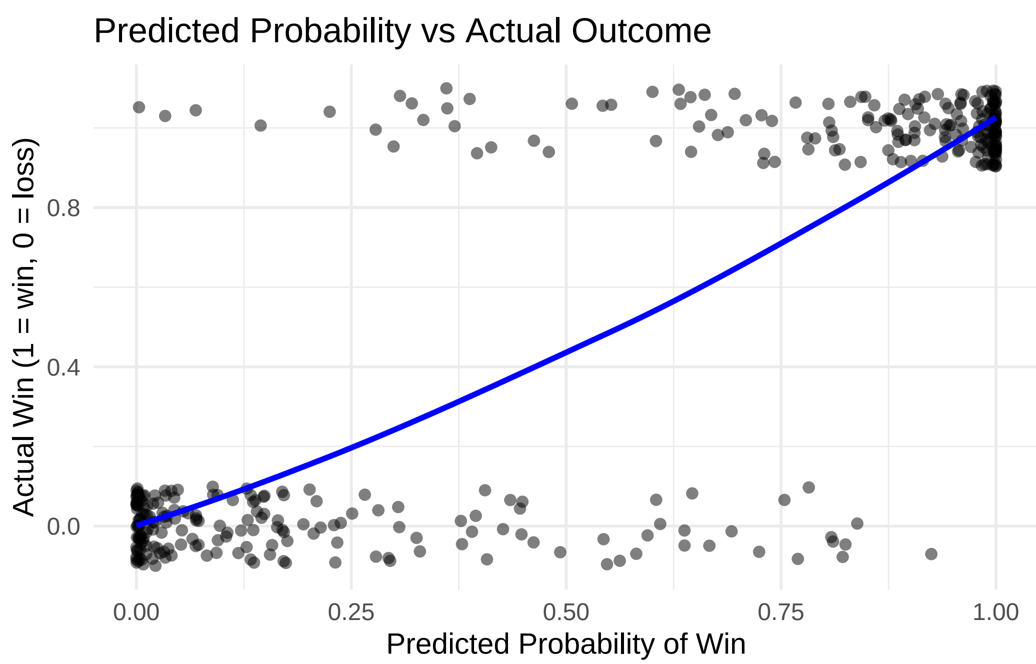
```
      aces_rate double_faults_rate     first_serve_pct     service_win_pct
       1.322372           1.235014            1.152589            1.953323
 return_win_pct
       1.567202
```

```
Confusion matrix for the test predictions of the initial model:

         Actual
Predicted   0   1
        0 162  18
        1  22 168

Test accuracy: 89.19 %

Sensitivity: 0.9032258

Specificity: 0.8804348
```

What may surprise in this results is that intuitively we may have expected aces_rate to have a strong (significant) impact, but the model disproves it.

The accuracy of the initial model is also quite high, 88.92%, as is the sensitivity, 90.66%, and specificity, 87.23%.

The plot for predictions based on the initial model:

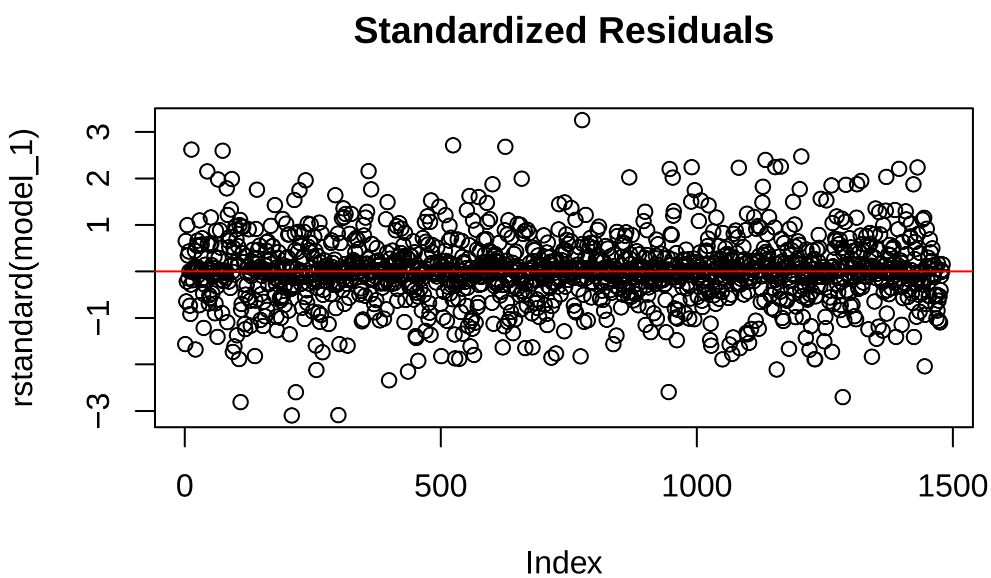


After seeing the results of the initial model we proposed the second one, which eliminated the statistically insignificant variables. This did not introduce great changes, the accuracy and the recall stayed pretty much the same, so we decided to go with the initial model.

```
Revised model summary



Call:
glm(formula = win ~ service_win_pct + return_win_pct, family = "binomial",
    data = train)

Coefficients:
                Estimate Std. Error z value Pr(>|z|)
(Intercept)      -30.066      1.696  -17.73   <2e-16 ***
service_win_pct   36.640      2.167   16.91   <2e-16 ***
return_win_pct    23.936      1.617   14.80   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 2051.71  on 1479  degrees of freedom
Residual deviance:  783.51  on 1477  degrees of freedom
AIC: 789.51

Number of Fisher Scoring iterations: 7

VIF values for the revisedl model:

service_win_pct  return_win_pct
       1.561251        1.561251

Comparing the AIC scores of the initial and revised models:

             df      AIC
model_1       6 791.0407
model_simple  3 789.5057

Confusion matrix for the test predictions of the revised model:
```

```
         Actual
Predicted   0   1
        0 160  21
        1  24 165

Test accuracy: 87.84 %

Sensitivity: 0.8870968

Specificity: 0.8695652
```

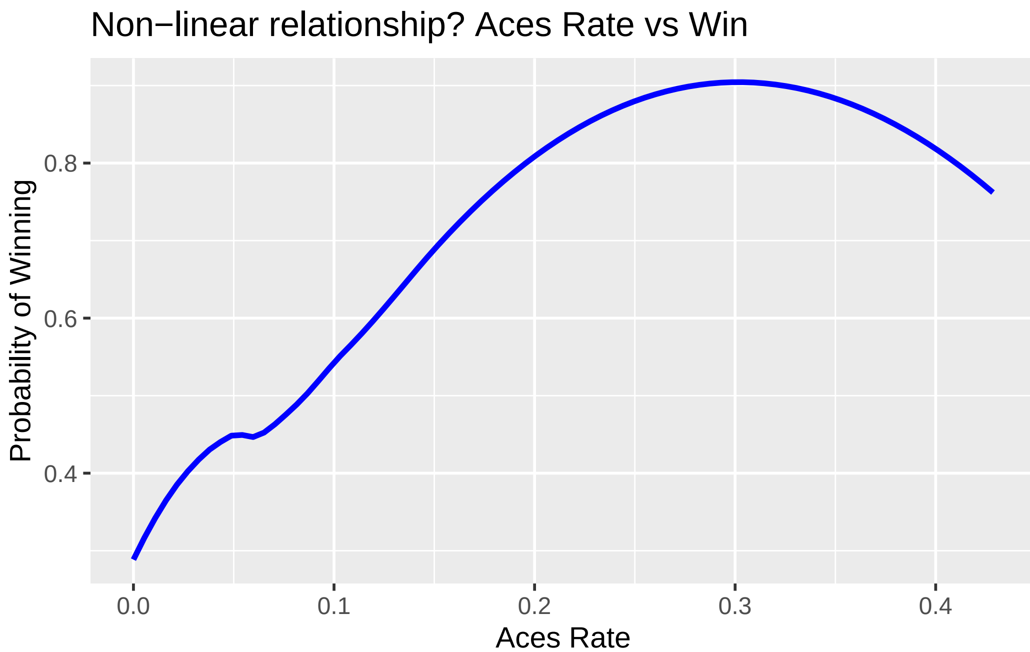
Also the 10-fold cross-validation showed the accuracy of   88%, which tells us that the results we get are not random and the model is pretty consistent.



The residuals plot showed no visible trend, which tells us that the linearity assumption is not violated, and no homoscedasticity was detected (insert residuals plot).

The correlation plot mentioned earlier showed moderate correlation between variables aces_rate and service_win_pct, but the multicollinearity check that was conducted on the model didn't show anything significant. The VIF values of the model all lie between 1 and 2, which is considered a good result and absence of the multicollinearity. (see results for initial model above)

Another thing we looked into was the presence of the non-linear dependence, but the check also didn't show anything significant.

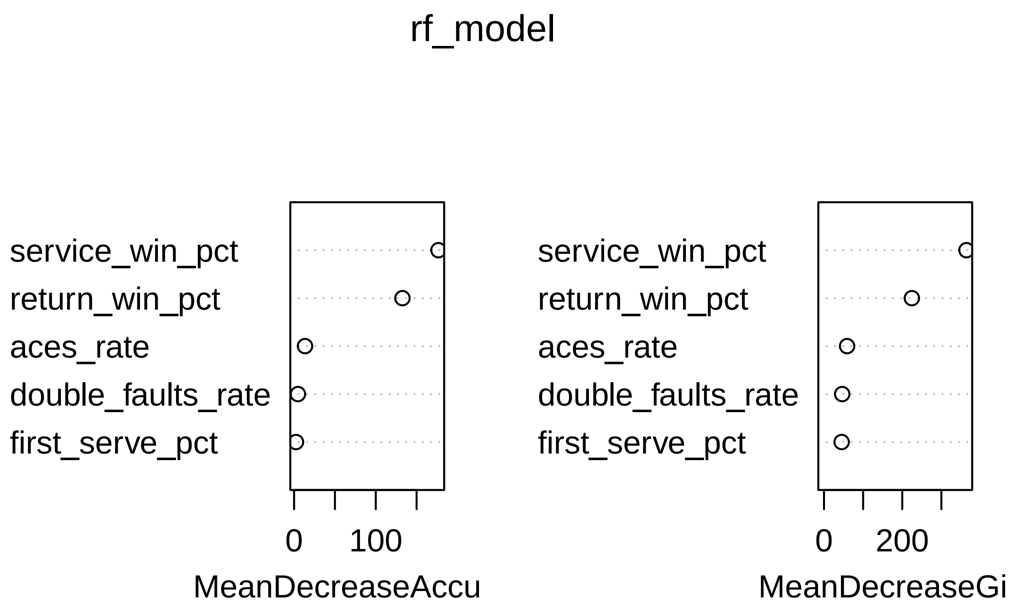



```
Call:
glm(formula = win ~ I(aces_rate^2), family = "binomial", data = train)

Coefficients:
               Estimate Std. Error z value Pr(>|z|)
(Intercept)    -0.39207    0.06736  -5.821 5.86e-09 ***
I(aces_rate^2) 46.35533    5.63244   8.230  < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 2051.7  on 1479  degrees of freedom
Residual deviance: 1949.3  on 1478  degrees of freedom
AIC: 1953.3

Number of Fisher Scoring iterations: 4


         Actual
```

```
Predicted   0   1
        0 142 120
        1  42  66

Test accuracy: 56.22 %

Sensitivity: 0.3548387

Specificity: 0.7717391
```

Additionally we decided to try the random forest approach, which confirmed the results of the initial model.

```
         Actual
Predicted   0   1
        0 164  24
        1  20 162

Random Forest Accuracy: 88.11 %

                             0          1 MeanDecreaseAccuracy MeanDecreaseGini
aces_rate           10.3548264   7.686753            13.352669         59.03101
double_faults_rate   2.8743402   3.556839             4.684929         46.51228
first_serve_pct      0.9781362   2.081076             2.320109         45.11852
service_win_pct    134.8654135 133.882252           176.745322        364.19279
return_win_pct      95.2232343 104.497855           132.642898        224.59981
```



Finally it is very important to say, that the model we chose performs well only on the given data and can not be generalized because of lack of history data and other important variables.

Additional Analyses

The performance of winning players varied notably by court type. On grass courts, winners recorded the highest percentage of service aces (0.11%) and service points won (0.71%), indicating grass courts favor stronger service games. Clay court winners, meanwhile, had the lowest service ace rate (0.07%) but performed best in first serve return won percentage (0.36%) and had competitive service points won (0.68%), highlighting the greater importance of return skills and extended rallies on clay. Hard court winners had the highest double faults percentage (0.035%). Overall, these results show performance of players differ across the three court surfaces.

Court Type	Service Aces(%)	Double Faults (%)	First Serves In (%)	Service Points Won (%)	First Serve Return Won (%)
Clay	0.07 a[2]	0.032 ab	0.62 a	0.68 a	0.36 a
Grass	0.11 b	0.031 a	0.64 b	0.71 b	0.32 b
Hard Court	0.09 c	0.035 b	0.61 a	0.69 a	0.34 a

[2]Means in the same column are significantly different at 5% level of significance if not followed by any letter in common.

Performance of Winners among Court Types

Among the top three-ranked men's singles players in 2017, Federer stood out with the highest percentage of service aces (0.138%) and service points won (0.73%), indicating his dominant and effective serve. Nadal excelled in first serves in (0.67%) and first serve return won (0.37%), reflecting his consistent serving and strong return game. Dimitrov, meanwhile, had the highest rate of double faults (0.042%) but maintained competitive figures in other areas. Compared to the averages, all three players exceeded the benchmarks for first serves in, service points won, and first serve return won, highlighting their all-around superiority in both offensive and defensive aspects of the game.

Rank	Player	Service Aces (%)	Double Faults (%)	First Serves In (%)	Service Points Won (%)	First Serve Return Won (%)
1	Nadal	0.049	0.023	0.67	0.71	0.37
2	Federer	0.138	0.020	0.64	0.73	0.34
3	Dimitrov	0.099	0.042	0.63	0.69	0.32
-	Average	0.067	0.041	0.61	0.63	0.29

Performance of the Highest Ranked Players (2017)

Conclusion

Summary

Winning players demonstrated consistently strong performance across all aspects of their game. Both their serving and returning abilities were above average when compared to losing players. In particular, winners secured significantly more points on their first serves as well as on returns against their opponents' first serves. These two metrics (service points won and first serve return points) emerged as reliable predictors of match success. Contrary to common intuition, the ace rate did not prove to be a particularly strong indicator of whether a player would win.

Limitations

This analysis is subject to several important limitations. First, the dataset includes only match-level data from a single year (2017) and covers just six ATP tournaments. Given that the ATP World Tour comprises over 60 tournaments annually, this selection represents only a small fraction of the full tour calendar. As a result, the findings may not be representative of player performance across the broader tour or over time.

Second, the limited time frame restricts the possibility of longitudinal analysis or year-over-year comparisons. Certain top-ranked players, such as Novak Djokovic, appear to be absent from the dataset, possibly due to injury or non-participation in the selected tournaments. Including data from all four Grand Slam tournaments over multiple years would likely allow for more comprehensive and generalizable insights.

Third, the dataset lacks several key performance metrics commonly used in tennis analysis, such as the number of winners, unforced errors, and break points won. While tiebreak information is included, it is less informative than break point data, which is often considered a critical indicator of match dynamics and player resilience under pressure.